

From Second Quantization and Quasispin to Pauli Strings

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Scope of these notes

These notes explain how to rewrite many-body Hamiltonians in terms of Pauli strings built from

$$I, \quad X, \quad Y, \quad Z,$$

so that they are ready for quantum-computing implementations.

Main topics:

- the Lipkin Hamiltonian in quasispin form,
- detailed mappings to Pauli operators,
- compressed and direct qubit encodings,
- second-quantized fermionic operators $a^\dagger, a$,
- the nuclear pairing Hamiltonian as a second worked example,
- Jordan–Wigner-type mappings from fermions to qubits.

Notation

We use:

- $a_p^\dagger$: fermion creation operator in orbital p ,
- a_p : fermion annihilation operator in orbital p ,
- $n_p = a_p^\dagger a_p$: number operator,
- X_i, Y_i, Z_i : Pauli operators on qubit i ,
- I_i : identity on qubit i .

Throughout, we avoid $c_p^\dagger, c_p$ notation and use only

$$a_p^\dagger, \quad a_p.$$

For quasispin systems we use collective operators

$$J_x, J_y, J_z, \quad J_\pm = J_x \pm iJ_y.$$

Why Pauli decompositions matter

Quantum computers natively implement unitary evolutions generated by sums of Pauli strings, for example

$$H = \sum_{\alpha} h_{\alpha} P_{\alpha}, \quad P_{\alpha} \in \{I, X, Y, Z\}^{\otimes n}.$$

Once a Hamiltonian is written in this form, one can:

- estimate energies in VQE,
- generate Trotterized time evolution,
- build QAOA circuits,
- analyze measurement grouping and commutativity.

So the essential preprocessing step is:

rewrite the many-body Hamiltonian as a weighted sum of Pauli strings.

We consider the Lipkin Hamiltonian in quasispin representation

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2.$$

Here:

- N is the number of particles,
- ϵ sets the one-body splitting,
- V controls pair scattering between the two levels,
- W weights the quadratic J_z^2 term.

The collective operators satisfy the $su(2)$ algebra

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y,$$

or, equivalently,

$$[J_z, J_{\pm}] = \pm J_{\pm}, \quad [J_+, J_-] = 2J_z.$$

The basis states are

$$|j, m\rangle, \quad m = -j, -j+1, \dots, j,$$

with

$$J_z |j, m\rangle = m |j, m\rangle.$$

Action of ladder operators

The ladder operators act as

$$J_+ |j, m\rangle = \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle,$$

$$J_- |j, m\rangle = \sqrt{j(j+1) - m(m-1)} |j, m-1\rangle.$$

Therefore

$$J_+^2 |j, m\rangle = \sqrt{j(j+1) - m(m+1)} \sqrt{j(j+1) - (m+1)(m+2)} |j, m+2\rangle,$$

and analogously for J_-^2 .

Useful identity for the interaction term

Using

$$J_{\pm} = J_x \pm iJ_y,$$

we compute

$$J_+^2 + J_-^2 = (J_x + iJ_y)^2 + (J_x - iJ_y)^2 = 2(J_x^2 - J_y^2).$$

Hence the Lipkin Hamiltonian can also be written as

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + V(J_x^2 - J_y^2) + \frac{W}{2} J_z^2.$$

This form is often the most convenient when collective spin operators are mapped directly to qubits.

Two mapping strategies

There are two main strategies for turning the Lipkin Hamiltonian into Pauli strings:

- 1 **Compressed encoding of the quasispin basis:** encode the $(2j + 1)$ -dimensional quasispin space into $\lceil \log_2(2j + 1) \rceil$ qubits.
- 2 **Direct collective-spin mapping:** write

$$J_\alpha = \frac{1}{2} \sum_i \sigma_i^\alpha$$

on a multi-qubit register and expand the Hamiltonian.

We will use:

- compressed encoding for $N = 4$,
- direct collective mapping for $N = 6$.

Four particles: quasispin dimension and encoding

For $N = 4$,

$$j = \frac{N}{2} = 2, \quad 2j + 1 = 5.$$

Thus the quasispin sector has dimension 5, which fits into 3 qubits since

$$2^2 = 4 < 5 \leq 8 = 2^3.$$

We choose the binary map

$$|2, -2\rangle \leftrightarrow |000\rangle, \quad |2, -1\rangle \leftrightarrow |001\rangle, \quad |2, 0\rangle \leftrightarrow |010\rangle,$$

$$|2, 1\rangle \leftrightarrow |011\rangle, \quad |2, 2\rangle \leftrightarrow |100\rangle.$$

The states $|101\rangle, |110\rangle, |111\rangle$ are unphysical.

Projectors and transition operators on qubits

For one qubit:

$$\begin{aligned} |0\rangle\langle 0| &= \frac{I + Z}{2}, & |1\rangle\langle 1| &= \frac{I - Z}{2}, \\ |1\rangle\langle 0| &= \frac{X - iY}{2}, & |0\rangle\langle 1| &= \frac{X + iY}{2}. \end{aligned}$$

Therefore:

- diagonal operators are expanded with I and Z ,
- off-diagonal transitions are expanded with X and Y ,
- any finite operator in the encoded space can be written as a sum of Pauli strings.

Three-qubit form of J_z

Since

$$J_z = \sum_{m=-2}^2 m |m\rangle \langle m|,$$

we obtain

$$J_z = -2 |000\rangle \langle 000| - |001\rangle \langle 001| + |011\rangle \langle 011| + 2 |100\rangle \langle 100|.$$

Expanding the projectors gives

$$J_z = -\frac{1}{4} IZI + \frac{1}{4} IZZ - \frac{1}{2} ZII - \frac{1}{2} ZIZ - \frac{3}{4} ZZI - \frac{1}{4} ZZZ.$$

Three-qubit form of J_z^2

Similarly,

$$J_z^2 = \sum_{m=-2}^2 m^2 |m\rangle \langle m|,$$

which yields

$$J_z^2 = \frac{5}{4} III + \frac{3}{4} IIZ + IZI + IZZ + \frac{1}{4} ZII - \frac{1}{4} ZIZ.$$

These formulas are exact on the physical 5-dimensional subspace.

Three-qubit form of the ladder terms

For $j = 2$,

$$\begin{aligned} J_+ |-2\rangle &= 2 |-1\rangle, & J_+ |-1\rangle &= \sqrt{6} |0\rangle, \\ J_+ |0\rangle &= \sqrt{6} |1\rangle, & J_+ |1\rangle &= 2 |2\rangle. \end{aligned}$$

Therefore

$$\begin{aligned} J_+^2 + J_-^2 &= \frac{3 + \sqrt{6}}{2} IXI + \frac{\sqrt{6} - 3}{2} IXZ + \frac{\sqrt{6}}{2} XXI + \frac{\sqrt{6}}{2} XXZ \\ &+ \frac{\sqrt{6}}{2} YYI + \frac{\sqrt{6}}{2} YYZ + \frac{3 + \sqrt{6}}{2} ZXI + \frac{\sqrt{6} - 3}{2} ZXZ. \end{aligned}$$

Final $N = 4$, 3-qubit Lipkin Hamiltonian

Substituting into

$$H = \frac{\epsilon}{2}(J_z - 2) + \frac{V}{2}(J_+^2 + J_-^2) + \frac{W}{2}J_z^2$$

gives

$$\begin{aligned} H_{N=4}^{(3q)} = & \left(-\epsilon + \frac{5W}{8}\right) III + \frac{3W}{8} IIZ + \left(-\frac{\epsilon}{8} + \frac{W}{2}\right) IZI + \left(\frac{\epsilon}{8} + \frac{W}{2}\right) IZZ \\ & + \left(-\frac{\epsilon}{4} + \frac{W}{8}\right) ZII + \left(-\frac{\epsilon}{4} - \frac{W}{8}\right) ZIZ - \frac{3\epsilon}{8} ZZI - \frac{\epsilon}{8} ZZZ \\ & + \frac{V(3 + \sqrt{6})}{4} IXI + \frac{V(\sqrt{6} - 3)}{4} IXZ + \frac{V\sqrt{6}}{4} XXI + \frac{V\sqrt{6}}{4} XXZ + \frac{V\sqrt{6}}{4} YYI + \frac{V\sqrt{6}}{4} YYZ + \frac{V(3 + \sqrt{6})}{4} ZII \\ & + \frac{V(\sqrt{6} - 3)}{4} ZIZ + \frac{V\sqrt{6}}{4} ZZI + \frac{V\sqrt{6}}{4} ZZZ \end{aligned}$$

Because the 3-qubit register has 8 basis states but only 5 are physical, one often adds

$$H_{\text{pen}} = \lambda P_{\text{unphys}},$$

where

$$P_{\text{unphys}} = |101\rangle \langle 101| + |110\rangle \langle 110| + |111\rangle \langle 111|.$$

This energetically suppresses leakage outside the encoded quasispin space.

Six particles: direct collective-spin realization

For $N = 6$, a simple and transparent mapping uses six qubits and defines

$$J_x = \frac{1}{2} \sum_{i=1}^6 X_i, \quad J_y = \frac{1}{2} \sum_{i=1}^6 Y_i, \quad J_z = \frac{1}{2} \sum_{i=1}^6 Z_i.$$

This is not qubit-optimal, since the symmetric quasispin sector has dimension 7, but it leads to a very clean Pauli-string Hamiltonian.

Expanding the collective squares

Using $X_i^2 = Y_i^2 = Z_i^2 = I$,

$$J_x^2 = \frac{1}{4} \sum_{i=1}^6 I + \frac{1}{4} \sum_{i \neq j} X_i X_j = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} X_i X_j,$$

and similarly

$$J_y^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Y_i Y_j,$$

$$J_z^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j.$$

Interaction term on 6 qubits

Using

$$J_+^2 + J_-^2 = 2(J_x^2 - J_y^2),$$

we get

$$\frac{V}{2}(J_+^2 + J_-^2) = V(J_x^2 - J_y^2) = \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j).$$

Thus the interaction is an all-to-all two-body Pauli interaction.

Linear and quadratic J_z terms on 6 qubits

For the linear term,

$$\frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) = \frac{\epsilon}{2} \left(\frac{1}{2} \sum_{i=1}^6 Z_i - 3I \right) = \frac{\epsilon}{4} \sum_{i=1}^6 Z_i - \frac{3\epsilon}{2} I.$$

For the quadratic term,

$$\frac{W}{2} J_z^2 = \frac{W}{2} \left(\frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j \right) = \frac{3W}{4} I + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

Final $N = 6$, 6-qubit Lipkin Hamiltonian

Collecting all terms gives

$$H_{N=6}^{(6q)} = \left(-\frac{3\epsilon}{2} + \frac{3W}{4} \right) I + \frac{\epsilon}{4} \sum_{i=1}^6 Z_i + \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j) + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

This is already directly usable in qubit-based quantum algorithms.

Comparison of the two Lipkin mappings

Case	Qubits	Main feature
$N = 4$, compressed	3	minimal encoding
$N = 6$, direct	6	simplest Pauli structure

Compressed mappings save qubits but require code-space control. Direct mappings use more qubits but give very transparent Hamiltonians.

Second quantization for fermions

In fermionic many-body theory, the basic operators are

$$a_p^\dagger, \quad a_p,$$

satisfying anticommutation relations

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = 0, \quad \{a_p^\dagger, a_q^\dagger\} = 0.$$

A general second-quantized Hamiltonian has the form

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} v_{pqrs} a_p^\dagger a_q^\dagger a_s a_r.$$

Why fermion-to-qubit mappings are needed

A quantum computer acts on qubits, not directly on fermionic Fock states. So we need a mapping

$$a_p^\dagger, a_p, n_p = a_p^\dagger a_p \longrightarrow \text{Pauli strings.}$$

The most common constructions are:

- Jordan–Wigner,
- Bravyi–Kitaev,
- parity mapping.

We focus here on Jordan–Wigner, since it is the most direct mathematically.

Jordan–Wigner transformation

For orbital p , the Jordan–Wigner mapping is

$$a_p^\dagger = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p - iY_p}{2}, \quad a_p = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p + iY_p}{2}.$$

The string of Z 's ensures the correct fermionic anticommutation relations.

Number operators under Jordan–Wigner

Using the mapping above,

$$n_p = a_p^\dagger a_p = \frac{1 - Z_p}{2}.$$

This is one of the simplest and most useful identities:

$$n_p \leftrightarrow \frac{1 - Z_p}{2}.$$

Thus one-body diagonal terms in fermionic Hamiltonians immediately become sums of I and Z .

Hopping terms under Jordan–Wigner

For $p < q$, one finds that

$$a_p^\dagger a_q + a_q^\dagger a_p$$

maps to strings of the form

$$\frac{1}{2} (X_p Z_{p+1} \cdots Z_{q-1} X_q + Y_p Z_{p+1} \cdots Z_{q-1} Y_q).$$

So hopping terms become nonlocal Pauli strings because of the fermionic parity chain.

Nuclear pairing Hamiltonian

A standard reduced pairing Hamiltonian is

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q,$$

where

$$N_p = a_p^\dagger a_p + a_{\bar{p}}^\dagger a_{\bar{p}},$$
$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p.$$

Here p and $\bar{p}$ denote time-reversed partners.

Pairing operators and quasispin structure

The pairing Hamiltonian is naturally associated with a quasispin algebra:

$$S_p^+ = P_p^\dagger, \quad S_p^- = P_p, \quad S_p^z = \frac{1}{2}(N_p - 1).$$

These satisfy

$$[S_p^+, S_q^-] = 2\delta_{pq}S_p^z, \quad [S_p^z, S_q^\pm] = \pm\delta_{pq}S_p^\pm.$$

Thus pairing and Lipkin Hamiltonians are closely related at the algebraic level: both can be expressed in terms of collective $su(2)$ -type generators.

One pair level: direct qubit mapping

For a single pair $(p, \bar{p})$, the two-dimensional paired subspace is naturally mapped to one qubit:

$$|0\rangle_p \equiv \text{pair unoccupied}, \quad |1\rangle_p \equiv \text{pair occupied}.$$

In this reduced pairing subspace, the quasispin operators become

$$S_p^+ = \frac{X_p - iY_p}{2}, \quad S_p^- = \frac{X_p + iY_p}{2}, \quad S_p^z = -\frac{Z_p}{2}.$$

Also,

$$N_p = 1 - Z_p.$$

Pairing Hamiltonian in pair-qubit form

Substituting

$$N_p = 1 - Z_p, \quad P_p^\dagger = \frac{X_p - iY_p}{2}, \quad P_p = \frac{X_p + iY_p}{2},$$

we obtain

$$H_{\text{pair}} = \sum_p \epsilon_p (1 - Z_p) - G \sum_{p,q} S_p^+ S_q^-.$$

Since

$$S_p^+ S_q^- + S_q^+ S_p^- = \frac{1}{2}(X_p X_q + Y_p Y_q),$$

the interaction becomes an XY -type coupling between pair qubits.

Explicit pairing Hamiltonian for Ω pair levels

For Ω pair orbitals, the pairing Hamiltonian in the paired subspace can be written as

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \epsilon_p (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q) - G \sum_p S_p^+ S_p^-.$$

Since

$$S_p^+ S_p^- = \frac{1 - Z_p}{2},$$

this yields

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \left(\epsilon_p - \frac{G}{2} \right) (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q).$$

This is already in Pauli-string form.

Example: two pair levels

For two pair orbitals $p = 1, 2$, the paired-subspace Hamiltonian becomes

$$H_{\text{pair}}^{(2)} = \left(\epsilon_1 - \frac{G}{2} \right) (1 - Z_1) + \left(\epsilon_2 - \frac{G}{2} \right) (1 - Z_2) - \frac{G}{2} (X_1 X_2 + Y_1 Y_2).$$

This is one of the simplest nontrivial pairing Hamiltonians ready for quantum simulation.

What if one uses full fermionic orbitals instead?

If one keeps both p and $\bar{p}$ as separate fermionic modes and applies Jordan–Wigner directly, then

$$a_p^\dagger a_{\bar{p}}^\dagger, \quad a_{\bar{p}} a_p$$

become longer Pauli strings with parity Z -chains.

This is fully general and exact in the full Fock space, but less compact. The pair-qubit reduction is much more efficient when seniority-zero paired states are the physically relevant sector.

Lipkin versus pairing: structural comparison

Both Hamiltonians admit spin-like descriptions:

- Lipkin:

$$H_{\text{Lipkin}} = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2,$$

- Pairing:

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q.$$

Both become sums of Pauli strings after:

- choosing an encoding,
- expressing collective or pair operators in terms of X, Y, Z .

General recipe: from second quantization to qubits

For a fermionic Hamiltonian:

- 1 write the Hamiltonian in terms of $a_p^\dagger, a_p$,
- 2 decide whether a reduced quasispin or pair-qubit subspace is appropriate,
- 3 otherwise apply Jordan–Wigner or a related mapping,
- 4 simplify number and pair operators,
- 5 collect the resulting Pauli strings.

For collective-spin models:

- 1 write the model in J_x, J_y, J_z or ladder-operator form,
- 2 choose compressed or direct qubit encoding,
- 3 expand diagonal and off-diagonal matrix elements into Pauli strings.

Summary

We updated the Lipkin notes by:

- adding more mathematical details,
- using $a^\dagger, a$ notation consistently,
- explaining both compressed and direct qubit mappings,
- adding the nuclear pairing Hamiltonian as a second detailed example,
- showing how second-quantized Hamiltonians become Pauli-string Hamiltonians ready for quantum computation.

The main conceptual bridge is:

many-body algebra $\longrightarrow$ spin/qubit algebra $\longrightarrow$ Pauli strings.

End

Questions?